Agoura Math Circle Curriculum

Class Schedule

Class	Quiz (5 min.)	Lecture	Class Work (30 min.)	Review (75 min.)	Homework
Junior Beginner	5 Questions	40 min.	15 Questions	75 min.	15 Questions
Junior Intermediate	5 Questions	40 min.	15 Questions	75 min.	15 Questions
Junior Advanced	5 Questions	40 min.	15 Questions	75 min.	15 Questions
Senior Beginner	5 Questions	40 min.	15 Questions	75 min.	15 Questions
Senior Intermediate	5 Questions	40 min.	15 Questions	75 min.	15 Questions
Senior Advanced	5 Questions	40 min.	15 Questions	75 min.	15 Questions

Merits:

1. At the end of each Semester (Fall, Spring) 2 students will receive the Star Award which is based on the cumulative score from the quiz, classwork, and homework.
2. Along with the Star Award, at the end of each Semester, 3 students will receive an award based on the final exam of that session.
3. New students will be placed in a class based on their assessment test which will take on the first day of the class.
4. Based on feedback from the instructor and the class coordinator and their performance in Final Exam, existing students will be able to move up to a higher class.

Class Material:

1. Students need to use the Message Center for communication with their instructor.
2. Parents need to help their kids if they are online students with homework, classwork, lecture notes, and the quiz.
3. All homework and classwork have bonus questions for extra credit.
4. The schedule is subject to change due to guest lectures and other special events.
5. All the class Lecture's videos available online in AMC YouTube channel. Students must watch Lecture's Video before coming to the class. YouTube channel subscription is required.
6. Registration/Enrollment is required for each semester (Fall and Spring). If kids take more than 2 class per semester, automatically system will drop them.

Junior Beginner - Elementary Math

Counting and Cardinality • Know number names and the count sequence. • Count to tell the number of objects. • Compare numbers. Operations and Algebraic Thinking • Understand addition as putting together and adding to and understand subtraction as taking apart and taking from. Number and Operations in Base Ten • Work with numbers 11–19 to gain foundations for place value. Measurement and Data • Describe and compare measurable attributes. • Classify objects and count the number of objects in categories. Geometry • Identify and describe shapes. • Analyze, compare, create, and compose shapes.

Session	**Agenda**
Fall 1	Counting: Counting Principles, Counting and Pattern, Problem Solving strategies
Fall 2	Addition: Up to Two-digit addition (with Regrouping), Solve problems with addition and with basic keywords sum of, altogether etc.
Fall 3	Subtraction: Up to Two-digit subtraction (With Regrouping), Solve problems with subtraction and with basic keywords difference, minus, fewer etc.
Fall 4	Multiplication: Up to two-digit Multiplication, Solve problems with Multiplication.
Fall 5	Division: Up to two-digit Division, Solve problems with Division.
Fall 6	Fraction: Single Digit simple fraction with addition, Subtraction and Multiplication
Fall 7	Decimal: Place value up to 2 digit and Single Digit Decimal with addition, Subtraction and Multiplication
Fall 8	Mixed Operations: Order of Operations with Parentheses, Order of Operations without Parentheses, Multiples of Ten.
Fall 9	Review for Final Exam
Fall 10	**Final Exam and Awards Ceremony**
Spring 1	Time: Telling time in 5minutes and hours, Quarter and half, Elapsed time, Solve problems with time with basic operation (Addition, Subtraction, Multiplication and Division)
Spring 2	Money: Identify the coins and their values, Count money, Solve problems with money
Spring 3	Measurements/Units: Measure lengths indirectly and by iterating length units (Inches, centimeter, feet)
Spring 4	Competition Preparation: Math Kangaroo International Competition for Level 1 & 2
Spring 5	Competition Preparation: Math Kangaroo International Competition for Level 1 & 2
Spring 6	Introduction to Geometry - Identify solid shapes Identify Faces, edges, vertices Basic shapes 2D and 3D. Classify objects and count the number of objects in categories. Finding the Area and perimeter for basic shapes (Square, rectangle) using single Digit.
Spring 7	Introduction to Probability - Basic Probability using coin, cards with single events.
Spring 8	Introduction to Statistics: Basic concept for Graph, Venn Diagram with single digit.
Spring 9	Introduction to Algebra: Basic Algebraic expressions using single digit
Spring 10	**Final Exam and AMC Fest with Year End Celebration**

Junior Intermediate – Intermediate Math

Add, subtract and solve problems involving addition and subtraction, use place value understanding and properties of operations to add and subtract. Work with time and money. Represent and solve problems involving multiplication and division up to 4 digits. Develop understanding of fractions as numbers with add, subtract, multiplication and Division, develop understanding of Decimal as numbers add, subtract, multiplication and Division, solve problems involving the four operations, and identify and explain patterns in arithmetic, solve problems involving measurement and estimation of intervals of time. Introduction to probability, graph, and geometric concepts.

Session	**Agenda**
Fall 1	Counting: Counting Principles, Counting and Pattern, Problem Solving strategies
Fall 2	Addition: Multistep Addition with large Number (with Regrouping), Solve problems with addition and with basic keywords sum of, altogether etc.
Fall 3	Subtraction: Multistep subtraction with large Number (With Regrouping), Solve problems with subtraction and with basic keywords difference, minus, fewer etc.
Fall 4	Multiplication: Multistep Multiplication with large Number and related Word problems.
Fall 5	Division: Multistep Division with large Number and related Word problems.
Fall 6	Fraction: Reciprocal, Improper Fractions, Mixed Fraction with Addition, Subtraction, Multiplication and Division.
Fall 7	Decimal: Place value for larger number, Larger Decimal Number with addition, Subtraction and Multiplication, and Division
Fall 8	Mixed Operations: Order of Operations (PEMDAS), Simplify Expressions Involving the Four Basic Operations and Problem Solving using PEMDAS.
Fall 9	Review for Final Exam
Fall 10	**Final Exam with Award ceremony**
Spring 1	Time & Money: Solve problems with Money with basic operation (Addition, Subtraction, Multiplication and Division) and Fraction/Decimal
Spring 2	Measurements/Units: Solve problems with measurement units within one system of units including km, m, cm, kg, g lbs. and oz.
Spring 3	Geometry Part 2- Identify solid shapes Identify Faces, edges, vertices Basic shapes 2D and 3D. Classify objects and count the number of objects in categories. Finding the Area and perimeter for basic shapes (Square, rectangle) using single Digit.
Spring 4	Competition Preparation: Math Kangaroo International Competition for Level 3 & 4
Spring 5	Competition Preparation: Math Kangaroo International Competition for Level 3 & 4
Spring 6	Problem Solving using by Probability - Basic Probability using coin, cards, dice with 2 events with solving problems.
Spring 7	Problem Solving using by Statistics: Basic concept for Graph, Venn Diagram with single digit.
Spring 8	Problem Solving using by Sets: Operations on sets, Symmetric Difference and Venn Diagrams.
Spring 9	Algebra: Basic Algebraic expressions using single digit
Spring 10	**Final Exam and AMC Fest with Year End Celebration**

Junior Advanced - Basic Pre - Algebra

4 operations with whole numbers Divisibility rule, Number and operations in Base ten Simplify problems to solve them by looking for patterns Pattern based problems, Factors and multiples LCM, GCF, Fractions common denominators ,Equivalence and ordering, Add, Subtract ,Multiply, Divide, Introduce larger numbers and operations, Fractions up to hundredths , Mix of common and uncommon denominator fractions and operations, Compare and convert fractions to decimals and vice-versa , Time based calculations, Basic algebraic expressions, Conversion from large to small unit Volume, relate to multiplication and addition, Graph points, coordinate plane to solve real-world problem, Study table of data , pictorial representation to answer questions, Angle, measure angles, lines, angles, shapes, Basic shapes based on lines and angles -- Perimeter, Area, Basic Probability.

Session	Agenda
Fall 1	Order of operations: Basic Operator's Properties, Order of operations (PEMDAS) and using the Order of Operations to Simplify Expressions Involving Addition, Subtraction, Multiplication, division.
Fall 2	Whole Number: Place value, Computation using basic operators, rounding off, Estimating.
Fall 3	Number Theory
Fall 4	Fractions: Reducing Fractions, Mixed Number with basic operators, Compound Fractions.
Fall 5	Fractions: LCM, GCF and Factoring based on Un Common Denominator.
Fall 6	Decimal: Place value, Comparing, Larger Decimal Number with addition, Subtraction and Multiplication, and Division, Estimation.
Fall 7	Percent: Change Fraction to decimal, Decimal to Fractions, changing decimal to percent, changing fractions to percent, Proportion Method, Percent –increase and decrease word problems.
Fall 8	Integers and Rational: Number Lines, Adding, Subtracting, Multiplying, and Dividing with integers. Absolute value. Adding, Subtracting, Multiplying, Dividing, Fraction, Decimal with positive and Negative integers.
Fall 9	Review for Final Exam
Fall 10	**Final Exam with Award ceremony**
Spring 1	Powers and Exponents: Powers, Exponents, Square and Cube, Simplify Square Roots. Operations with Powers and Exponents.
Spring 2	Scientific Notation: Powers of Ten, Multiplying and Dividing Powers of Ten and Scientific notation.
Spring 3	Measurements/Units: Solve problems with measurement units within Multiple system of units including km, m, cm, kg, g lbs. and oz.
Spring 4	Competition Preparation: Math Kangaroo International Competition for Level 5 & 6
Spring 5	Competition Preparation: Math Kangaroo International Competition for Level 5 & 6
Spring 6	Geometry - Measure angles in whole-number degrees using a protractor, Finding the Area and perimeter for basic shapes (Square, rectangle, and circle, etc.).
Spring 7	Probability - Probability using coin, cards, dice with Independent events.
Spring 8	Statistics: Problem solving using Venn Diagram, Bar Charts, Circle Charts, Line Graphs.
Spring 9	Algebra: Algebra Expressions, Multi variable and simple Equations.
Spring 10	**Final Exam and AMC Fest with Year End Celebration**

Senior Beginner - Pre-Algebra

Properties of Arithmetic, Exponents, Number Theory, Fractions, Equations and Inequalities, Decimals Ratios, Conversions, and Rates, Percent's, Square Roots, Angles, Perimeter and Area, Right Triangles and Quadrilaterals, Data and Statistics, Counting, Problem-Solving Strategies and Introduction to Algebra I & II

Session	**Agenda**
Fall 1	Problem Solving using by Order of Operations
Fall 2	Problem Solving using by Properties of Arithmetic
Fall 3	Problem Solving using by Exponents
Fall 4	Problem Solving using by Number Theory
Fall 5	Problem Solving using by Fractions
Fall 6	Problem Solving using by Equations and Inequalities
Fall 7	Problem Solving using by Ratios, Conversions, and Rates
Fall 8	Problem Solving using by Square Roots
Fall 9	Review for Final Exam
Fall 10	**Review and Final Exam with Award ceremony**
Spring 1	Problem Solving using by Geometry Part 1: Angles
Spring 2	Problem Solving using by Geometry Part 2: Perimeter and Area
Spring 3	Problem Solving using by Geometry Part 3: Right Triangles and Quadrilaterals
Spring 4	Competition Preparation: Math Kangaroo International Competition for Level 7 & 8
Spring 5	Competition Preparation: Math Kangaroo International Competition for Level 7 & 8
Spring 6	Problem Solving using by Data and Statistics
Spring 7	Problem Solving using by Counting
Spring 8	Introduction to Algebra I
Spring 8	Introduction to Algebra II
Spring 10	**Final Exam and AMC Fest with Year End Celebration**

Senior Intermediate - Basic Pre-Calculus

Algebra - Variables and Equations, Linear Equations, Systems of Equations, Ratios, Proportions Percent, Distribution, Quadratics, Exponents and Radicals, Statistics, Sequences and Series, Substitution.

Counting - Simple Counting Techniques, Venn Diagrams, Triangular Number Patterns, The Fundamental Counting Principle, Factorials, Permutations, Combinations, Pascal's Triangle.

Probability -Probability Basics, Compound Events and Counting, Casework and Probability, Probability and Combinations, Complementary Counting and Probability, Geometric Probability.

Number Theory - Primes and Divisibility, Factors, Factor Tricks, Different Bases, the Units Digit, Fractions and Decimals, Modular Arithmetic.

Geometry - Geometry Basics, Circles, Chords, Secants, Tangents, Arcs, Inscribed Angles, Properties, Circumference and Arc Length, Pythagorean Theorem, Area, Heron's Formula, Altitudes, Working Backwards, Three-Dimensional Geometry, Polyhedral, Faces, Vertices, and Edges, Similarity, Right Triangles, Circles, Parallel Lines, Area and Volume Relationships.

Precalculus: Introduction to Exponents and Logarithms, Trigonometry,

Session	**Agenda**
Fall 1	Algebra - Variables and Equations, Linear Equations, Systems of Equations
Fall 2	Algebra - Ratios, Proportions Percent, Distribution, Quadratics, Exponents and Radicals
Fall 3	Algebra - Statistics, Sequences and Series, Substitution
Fall 4	Counting - Simple Counting Techniques, Venn Diagrams, Triangular Number Patterns
Fall 5	Counting - The Fundamental Counting Principle, Factorials, Permutations
Fall 6	Counting - Combinations, Pascal's Triangle.
Fall 7	Probability -Probability Basics, Compound Events and Counting, Casework and Probability
Fall 8	Probability -Probability and Combinations, Complementary Counting and Probability
Fall 9	Competition Preparation: AMC 10
Fall 10	**Final Exam with Award ceremony**
Spring 1	Probability - Geometric Probability.
Spring 2	Number Theory - Primes and Divisibility, Factors, Factor Tricks.
Spring 3	Number Theory - Different Bases, Fractions and Decimals, Modular Arithmetic.
Spring 4	Competition Preparation: Math Kangaroo International Competition for Level 9 & 10
Spring 5	Competition Preparation: Math Kangaroo International Competition for Level 9 & 10
Spring 6	Geometry Part 1- Geometry Basics, Circles, Chords, Secants, Tangents, Arcs, Inscribed Angles, Properties, Circumference and Arc Length,
Spring 7	Geometry Part 2 - Pythagorean Theorem, Area, Heron's Formula, Altitudes. Working Backwards, Three-Dimensional Geometry, Polyhedral. Faces, Vertices, and Edges, Similarity, Right Triangles, Circles, Parallel Lines, Area and Volume Relationships.
Spring 8	Introduction to Exponents and Logarithms
Spring 9	Introduction to Trigonometry
Spring 10	**Final Exam and AMC Fest with Year End Celebration**

Senior Advanced

Pre- Calculus

Exponents and Logarithms, Trigonometry, Cyclic Quadrilaterals, Conics and Polar Coordinates, Polynomials, Functions, Limit, Complex Numbers, Vectors and Matrices, Reflection, Distortion, Dilation, Potpourri of Geometry, Binomial Theorem Three-Dimensional Geometry, Analytic Geometry, Equations and Expressions, combinatorics, Sequences and Series, Statistics and Probability, Number Theory, Diophantine Equations, How to Solve 3D Problems.

Calculus

Limits, Basics of Derivatives, Derivative Rules, Real Life Derivatives, Chain Rule, Maximum and Minimums, Related Rates, Derivative Review, Finding the Antiderivative, Integral Rules, Finding Indefinite Integrals, Finding Integrals using Rectangles, Reverse Chain Rule, Finding Definite Integrals, Integrals with Substitution, Finding the Area between Curves, Summation and Finding the Integral using Summation Notation, Review Summation Notation First.

Session	**Agenda**
Fall 1	Pre-Calculus: Exponents and Logarithms
Fall 2	Pre-Calculus: Trigonometry Part 1 – Introduction and Basic Concept.
Fall 3	Pre-Calculus: Trigonometry Part 2 – Advanced Concept.
Fall 4	Pre-Calculus: Complex Numbers
Fall 5	Pre-Calculus: Combinatorics
Fall 6	Pre-Calculus: Statistics and Probability
Fall 7	Pre-Calculus: Functions and Inequalities
Fall 8	Pre-Calculus: Sequences and Series
Fall 9	Competition Preparation: AMC 12
Fall 10	**Final Exam with Award ceremony**
Spring 1	Calculus - Limits
Spring 2	Calculus - Basics of Derivatives
Spring 3	Calculus - Real Life Applications of Derivatives
Spring 4	Calculus - Review Real Life Applications of Derivatives
Spring 5	Calculus - Finding the Antiderivative
Spring 6	Calculus - Finding Integrals using Rectangles
Spring 7	Calculus - Finding Definite Integrals
Spring 8	Calculus - Finding the Area between Curves
Spring 9	Calculus - Summation and Finding the Integral using Summation Notation
Spring 10	**Final Exam and AMC Fest with Year End Celebration**